

Riccardo Gozzo

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EDUCATION

Scuola Normale Superiore

Pisa, ITA

Ph.D in Computational Methods and Mathematical Models for Finance

Nov. 2025 – 2029

- **Main Courses:** Advanced quantitative finance, Statistical and machine learning, Numerical methods for SDE, Market microstructure in finance

University of Milano-Bicocca

Milan, ITA

Master's Degree in Statistical Sciences (Grade: 110L/110)

Oct. 2023 – Oct. 2025

- **Thesis & Working Paper:** Adaptive Multilevel Splitting: First Application to Rare-Event Derivatives
- **Software (CRAN):** Developed and maintained [**amsSim**], an R package implementing **AMS methods** for rare-event derivative pricing.
- **Main Courses:** Stochastic calculus for finance, Applied probability, Machine learning, Computational statistics, Computational finance, Optimal Control, Bayesian statistics

University of Milano-Bicocca

Milan, ITA

Bachelor's Degree in Statistical Sciences (Grade: 110/110)

Oct. 2020 – Jul. 2023

- **Main Courses:** Analysis (I, II), Linear algebra, Statistics (I, II, III), Probability theory, Econometrics (I, II), Data Mining, State space models

EXPERIENCE

BKN301

Milan, ITA

Student Research (Internship)

Mar. 2023 – Jun. 2023

- Developed a multi-currency collateralized stablecoin using algorithms (**Quadratic programming**, **GARCH**) and machine learning (**LSTM**) techniques to forecast and minimize variance, outperforming the benchmark by 16%
- Analyzed and back tested over 25 years of data and extended the idea to five additional types of collateral, reducing algorithmic operational costs by 3%

FINECO SPA

Rovigo, ITA

Curricular Internship

Sep. 2018 – Jun. 2019

- Studied and calculated management costs for 10 investment portfolios and created a report on Mifid 2 regulations

PROJECTS

Latent Graph Neural Networks for Volatility Forecasting

- Developed a **Latent Soft-Sparse GNN** with **Residual Gated Graph Convolutions** to endogenize financial network topology via a **differentiable graph constructor**. Captured non-linear spillovers from high-frequency data and formulated a **hybrid QLIKE loss** anchored to **HAR priors**, regularizing neural optimization and outperforming linear benchmarks by 15%.

Deep Learning for Portfolio Optimization

- Developed an optimization framework via **temporal neural networks** (LSTM, TSMixer). Mitigated Sharpe Ratio instability and gradient greedy bias using a **custom linear utility loss**, mathematically regularizing weight allocation under non-stationary dynamics and reducing benchmark oscillations by 50%.

Optiver: Trading at the close

- Implemented **ensemble tree-based models** (XGBoost, LightGBM) to forecast high-frequency variables during the Nasdaq closing auction. Addressed market microstructure noise by optimizing the objective function for **robust variance reduction**, outperforming the benchmark by reducing forecast error by 8%.

TECHNICAL SKILLS

Programming: Python, C++, R, MYSQL

Deep Learning Frameworks: PyTorch, TensorFlow

Languages: Italian (Native), English (Advanced C1)